

Project 2: Preparing Stock Data for Machine Learning

This report answers the questions in *Project 2: Preparing Stock Data for Machine Learning* using figures taken directly from notebook output.

The dataset is **historical_stock_prices.csv** (20,973,889 rows, 8 columns) joined to **historical_stocks.csv** (6,460 rows, 5 columns) on ticker, covering 2 January 1970 to 24 August 2018.

PART 2 QUESTIONS

- 1. What does **daily_return** tell us?** It is the percentage change in closing price from one trading day to the next. Unlike **price_change**, it is scale-free: a \$2 move in a \$20 stock and a \$200 move in a \$2,000 stock both register as 10%. That makes returns comparable across securities and across time, which is why almost every financial model is built on returns rather than on price levels.
- 2. What does a **moving average** tell us about a stock?** It smooths day-to-day noise to reveal the underlying trend. A short window (MA 5) responds quickly and follows price closely; a long window (MA 20) lags but is steadier. The relationship between them carries the signal, price above its moving average suggests upward momentum, and a short average crossing above a long one is a classic trend-change indicator.
- 3. What does **high volatility** indicate?** Large and unpredictable daily swings. It means greater uncertainty and risk, and it typically coincides with news, earnings, or broader market stress. It says nothing about direction, a stock can be highly volatile while trending up, down, or sideways. Volatility also clusters in time, so a high reading today raises the odds of a high reading tomorrow.
- 4. Why do lag and rolling features create missing values at the start of each stock's history?** Because they need prior observations that do not exist yet. **close_lag_1** on a ticker's first trading day has no previous day to reference. A 20-day rolling standard deviation cannot be computed until 20 observations have accumulated. Since the features are computed per ticker with **groupby**, the warm-up repeats at the start of every security rather than only once at the top of the file, which is correct, and is exactly what prevents one stock's history bleeding into another's.

PART 3 QUESTION

Why is one-hot encoding preferable to NASDAQ = 1, NYSE = 2? Integer codes impose an order and a distance that does not exist. The model would read NYSE as "greater than" NASDAQ, and with three or more categories it would infer that category 3 is as far from category 1 as category 1 is from category -1. Linear models would fit a coefficient to that fictional scale, and distance-based methods would compute meaningless distances from it. One-hot gives each category its own binary column, so no ordering is implied and the model learns an independent effect for each. `drop_first=True` removes the redundant column to avoid the dummy-variable trap in linear models.

PART 5 SUMMARY

- 1. What data-quality issues did you investigate?** Missing values in both files (none in prices; 1,440 rows missing sector and industry in the metadata); a header row that had been read in as a data record (`ticker = 'Symbol'`); OHLC arithmetic violations (48 rows with low > high, 446 with close > high, 230 with close < low, 965 with open > high, 931 with open < low); `adj_close` values implausibly far from close (46,432 rows); extreme single-day price moves above 500% (1,971 rows); ticker formatting; duplicate records; and calendar sanity, weekend and future-dated rows, both of which came back clean at zero.
- 2. What decisions did you make about unusual observations?** Flagged rather than deleted, in most cases. Outliers were identified two ways, per-ticker IQR (231,882 rows) and a per-ticker modified z-score using MAD (1,061,958 rows, 5.06%). Both were stored as boolean columns so that Project 3 can decide. The reasoning is that a stock genuinely can move 40% on an earnings surprise; flagging that as "unusual" is informative but removing it would delete the most important events in the dataset. The single header-artefact row was dropped outright since it is unambiguously not data. Rows with contradictory OHLC arithmetic were dropped, because the data gives no basis for deciding which field is wrong.
- 3. What new features did you create?** Required: `price_change`, `daily_return`, `close_lag_1`, `ma_5`, `ma_20`, `volatility_20`. Chosen additional feature: `hl_range_pct`. Beyond the requirement I also built multi-horizon returns (5- and 21-day), log returns, longer moving averages and their ratios to price, a volatility regime ratio, volume features on a log scale, and three technical indicators, RSI(14), Bollinger band position and MACD. Calendar features were encoded cyclically with sine/cosine pairs so that December sits adjacent to January.
- 4. Which feature may be most useful, and why?** `volatility_20`. Returns themselves are close to unpredictable, the correlation heatmap shows almost no linear relationship between any feature and next-day return. Volatility, by contrast, is strongly autocorrelated: today's reading is a genuinely good predictor of tomorrow's. So even if the direction of the next

move is hard to call, its likely *size* is forecastable, and that is what risk sizing and option pricing depend on. **ma_cross** is the runner-up as a directional signal.

5. **What does the final prepared dataset contain?** **stock_prepared.csv** holds the 2010s decade of daily records with raw OHLCV columns, the seven engineered features, three metadata columns (**exchange**, **sector**, **industry**), the ticker and date identifiers, and the target **target_next_return**. Warm-up rows from the rolling windows are removed, so every row is complete. Alongside it, **data/processed/** holds the chronological train / validation / test splits, the fitted preprocessor, and a manifest recording the cutoff dates and column lists.
6. **What additional information would help a predictive model?** Corporate-action history, split and dividend dates would let **adj_close** be reconciled properly instead of flagged. Market-wide context: an index return such as the S&P 500 would let the model separate a stock's idiosyncratic move from a move the whole market made. Fundamentals (earnings dates, P/E, market cap) would explain jumps that price history alone cannot. Short interest and bid-ask spreads would add a liquidity dimension. And a sentiment or news-volume signal would help with precisely the days the current feature set handles worst, the gaps and spikes.

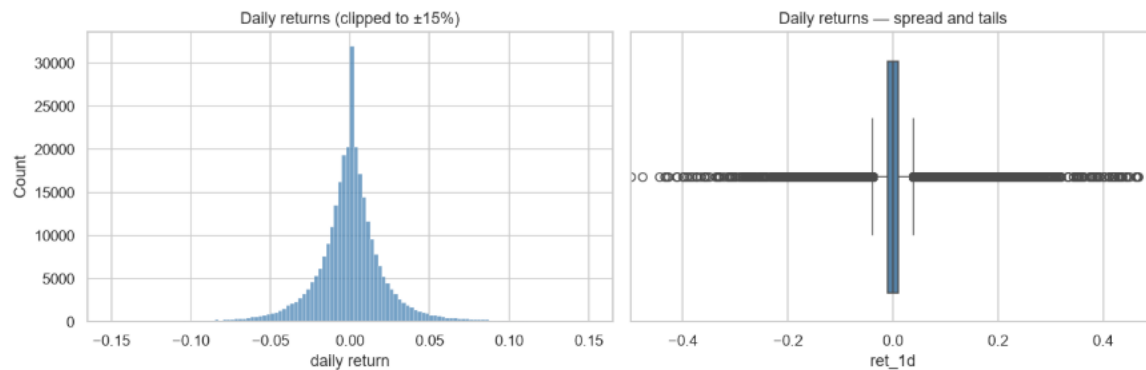
VISUALIZATIONS:

Distribution of daily returns:

Histogram (left): Most days, stocks move very little. The tall spike at 0 and the narrow bell shape mean the typical day is somewhere between -2% and +2%. The chart is clipped to $\pm 15\%$, so it hides the extreme values.

Boxplot (right): The box is so thin you can barely see it, because 50% of all days fall between about -1% and +1%. The long trail of circles on both sides are outliers, days with unusually big moves. Stock returns always have "fat tails" like this, so some of these are real (earnings surprises, crashes).

Bottom line: The median and quartiles are fine, but the **mean and std are broken by a small number of bad rows**. If you train a model on this as-is, those rows will distort it.



Volatility over time: The line chart shows how "jumpy" each stock was over time. Each line is the volatility (how much the price bounced around) over the last 21 trading days, about one month.

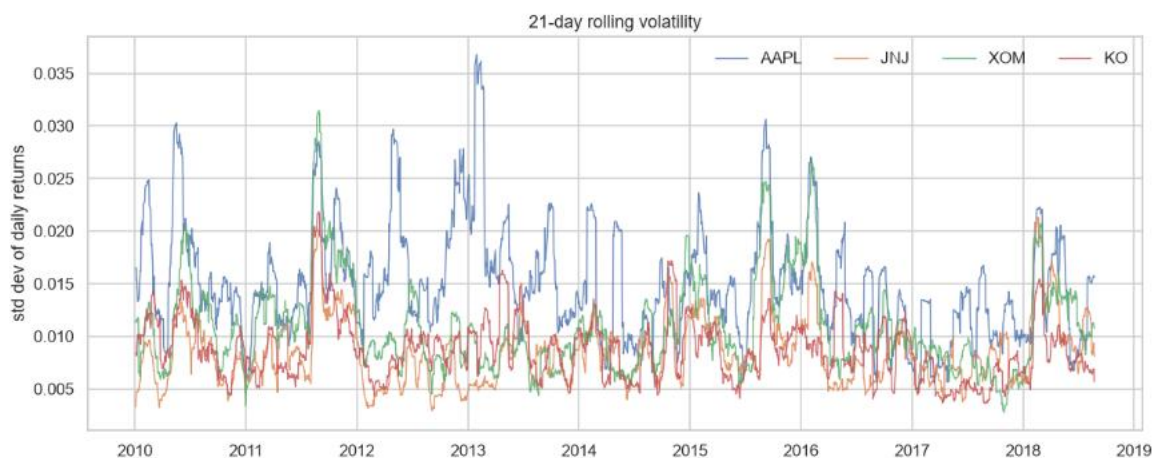
Higher line = bigger daily price swings = riskier. Lower line = calmer, steadier stock.

AAPL (blue) is the most volatile. It's usually on top, with the biggest spike in early 2013 (about 3.5% daily swings) when Apple's stock dropped sharply after earnings. Tech stocks tend to move more.

JNJ (orange) and KO (red) are the calmest. Johnson & Johnson and Coca-Cola usually stay around 0.5–1%. People buy things like medicine and soda no matter what the economy does, so these are called "defensive" stocks.

XOM (green) is in the middle, but it spikes when oil prices crash, like in late 2015 and early 2016.

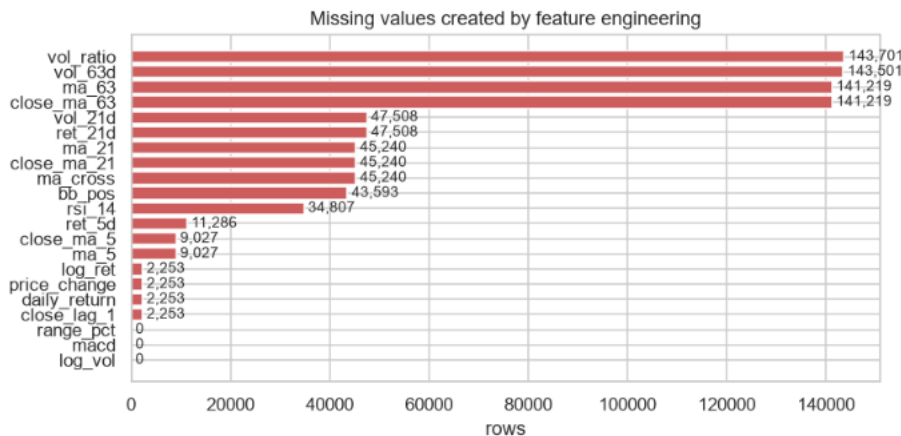
Volatility comes in waves. Calm periods and stressful periods cluster together, and risky events hit every stock at the same time, even the safe ones. That's useful for your model, because recent volatility (vol_21d) helps predict near-future volatility.



Missing Values: the bar chart shows how many empty (NaN) cells each new feature created. It's normal and expected, not a bug.

Many features need past days to calculate. A 21-day moving average can't be calculated on day 1 of a stock, because there aren't 21 days of history yet. So those first days are left empty.

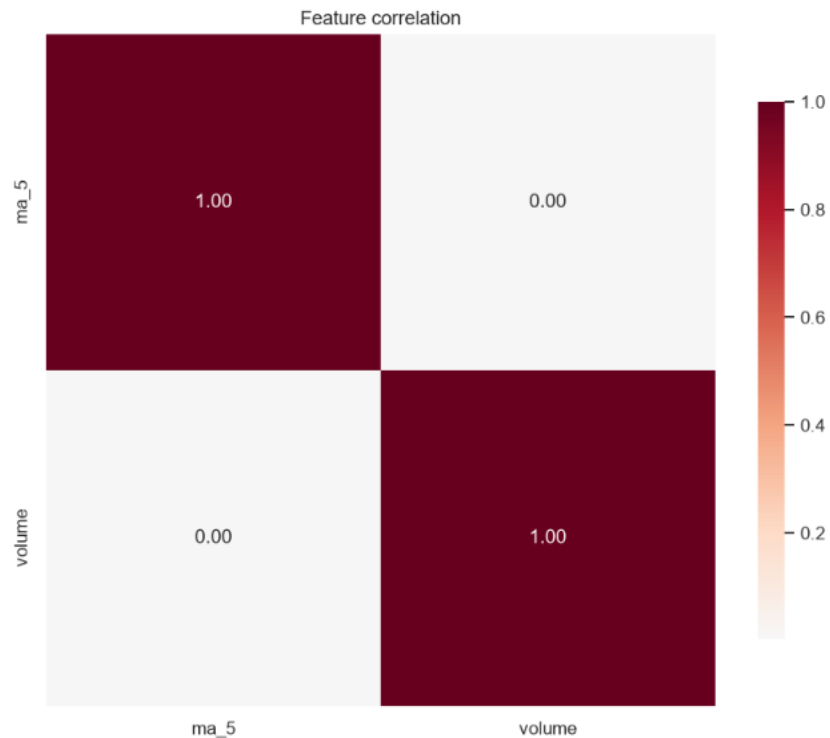
The pattern matches perfectly: the data has about **2,253 stocks**, and each stock loses its first few rows:



Feature Correlation: This heatmap shows how strongly features move together. 1.00 means they move perfectly together, 0.00 means no relationship.

What it shows:

- The dark squares on the diagonal are always 1.00, because every feature is perfectly related to itself. Ignore those.
- ma_5 and volume have 0.00: a stock's price level tells you nothing about how much it's traded. That makes sense, since a \$500 stock and a \$5 stock can both be heavily or lightly traded.



The split, drawn: The train (blue), dates 2010- 2014, rows 4.67M, share ~51%, purpose the model learns from this.

Validation (orange), 2015- 2016, 2.36M rows, share ~26%, Used to tune and compare models.

Test (green), 2017 – mid 2018, 2.15M, share ~23%, Final check, used only once at the end.

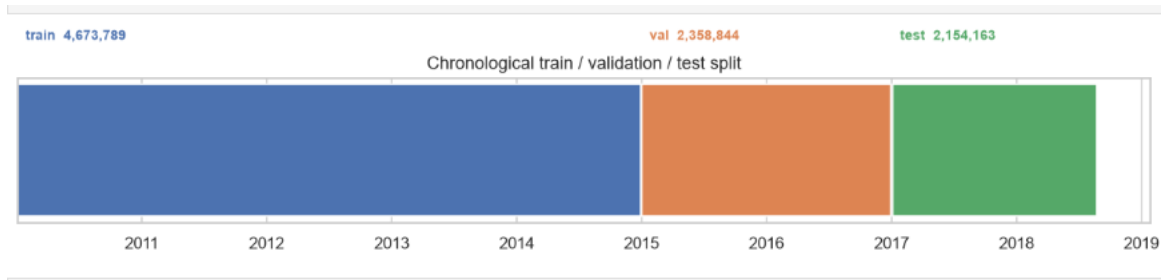
Split by time: With stock data you must predict the **future from the past**. If the split randomly, the model could train on 2017 data and then be "tested" on 2014, which is like seeing tomorrow's newspaper. It would look great in testing but fail in real life. This is called **data leakage**, and your chronological split avoids it.

The three parts don't overlap, and the order is correct: train → validation → test. This is the right way to do it.

Small things worth noticing:

- The test period is shorter (about 1.6 years) but still has many rows, because more stocks are in the data in later years.
- The market in 2017 was very calm, so test results may look a bit different from validation results. That's normal, not a mistake.
- If the target is **next-day** return, the last day of train uses the first day of validation as its answer. It's only one day per stock, so it barely matters, but some people drop a small gap at each boundary to be extra safe.

Note: the split is set up correctly.



DELIVERABLE CHECKLIST

- Project2_Ayleen.ipynb = notebook with code, comments, decisions, figures, answers
- stock_prepared.csv = Cell 11
- Project2_Ayleen.pdf = technical report
- All seven figures saved as PNG for the report